Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/02/2008
PI Name:	Hyman, Linda Ph.D.		
Project Title:	Genomic & Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique to Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02215	**Congressional District:**	8
Comments:	NOTE: PI formerly at Montana State University; move to Boston University effective 7/1/2009 (6/29/09)		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Nielsen-Preiss, Sheila (Montana State University)		
Grant/Contract No.:	NNJ04HC74G		
Performance Goal No.:			
Performance Goal Text:			
	Our experiments focus on understanding how cells adapt to the unique conditions of microgravity in the space, using the model eukaryotic organism, Saccharomyces cerevisiae. We will utilize two types of yeast microarray analysis, RNA 'expression' profiling and 'fitness' profiling. One set of experiments is designed to establish gene expression in the rotary suspension culture system that models microgravity conditions similar to the space environment. Once we identify genes regulating growth under selective pressure, we will confirm and extend our experiments, conducting parallel experiments with the yeast deletion series. Utilization of the yeast deletion series allows a genomic approach to phenotypic analysis of every strain representing the ~6000 genes in the yeast genome. Our hypothesis is that yeast cells will undergo a selective response to microgravity and that phenotypic responses are controlled by changes in gene expression of specific target genes. We expect this novel approach of comparing expression and fitness profiling datasets		

	to confirm some genes revealed by RNA expression analysis and also to reveal new genes required for growth that are not necessarily regulated transcriptionally.
Task Description:	Task Objectives 1. Identify yeast gene expression under selective pressure of low shear modeled microgravity in the HARV (completed, Years 1 and 2). 2. Confirm and extend preliminary work that identified specific promoter motifs that respond to changes in gene expression during growth in the HARV (ongoing). 3. Utilize a powerful genome-wide approach for analysis of highly specific alterations in eukaryotic cells attributable to microgravity (to be undertaken in Year 3). 4. Gain an understanding of biological mechanisms that may increase the health risks to humans during long-duration space flight (additional focus that addresses new critical path roadmap issues). Our research objectives can provide answers to questions addressing risks for human space flight. First, we can address whether microgravity exerts any selective pressures upon yeast cells. Second, we can answer whether space flight alters microbial growth rates and/or cell characteristics. Third, our research will provide insights into the molecular and genetic mechanisms that are affected by space flight related environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Regulatory mechanisms are largely conserved between yeast and mammalian cells. The ease of genetic manipulation, genome sequence information, and readily available commercial reagents in yeast facilitates the analysis of phenotypic and genotypic changes under selective growth conditions. Our expectation is that any biological effects observed in yeast cells in space will be similar to effects occurring in eukaryotic, mammalian cells. We hope genetic analysis will give clues to signaling mechanisms responsible for changes in gene expression. Any biological effects we identify may have direct implications for human space flight and long-term space missions. We hope our studies will establish methods for fitness profiling for other earth-based studies of stress responses. Furthermore, we believe that our phenotypic analysis results in Year 2 provide a platform for understanding developmental processes of higher eukaryotes that are known to employ spatially ordered and/or asymmetric cell division patterns. In addition, we have identified key regulatory genes responsible for biofim development in yeast that are differentially expressed in simulated microgravity. Further exploration of these results and uncovering the genetic and physiological factors affecting processes, will have important fundamental and medical implications.
Task Progress:	The final report for our grant describes the results that appear in four published manuscripts. The first discusses the effect of simulated microgravity in the yeast Saccahryomes cerevisiae demonstrating a clear phenotypic response to these growth conditions. In addition we show the molecular changes that accompany the morphological response we observe. The second presents the genomic response to growth in simulated microgravity which confims and extends out earlier observation. In a third and most recent paper we show that the response observed in Saccharomyces is conserved in the pathogenetic yeast, Candida albicans, which has significant implications for space flight. Future experiments will focus on this observation, particualry as it relates to increased pathogenesis and biofilm formation as the microorganism is exposed to microgravity conditions.
Bibliography Type:	Description: (Last Updated: 07/02/2008)
Articles in Peer-reviewed Journals	Altenburg SD, Nielsen-Preiss SM, Hyman LE. "Increased filamentous growth of Candida albicans in simulated microgravity." Genomics Proteomics Bioinformatics. 2008 Feb;6(1):42-50. PMID: 18558384 , Feb-2008
Articles in Peer-reviewed Journals	Purevdorj-Gage B, Orr ME, Stoodley P, Sheehan KB, Hyman LE. "The role of FLO11 in Saccharomyces cerevisiae biofilm development in a laboratory based flow-cell system." FEMS Yeast Res. 2007 May;7(3):372-9. Epub 2007 Jan 19. PMID: 17233763 , May-2007
Articles in Peer-reviewed Journals	Sheehan KB, McInnerney K, Purevdorj-Gage B, Altenburg SD, Hyman LE. "Yeast genomic expression patterns in response to low-shear modeled microgravity." BMC Genomics. 2007 Jan 3;8:3. PMID: 17201921 , Jan-2007
Articles in Peer-reviewed Journals	Purevdorj-Gage B, Sheehan KB, Hyman LE. "Effects of low-shear modeled microgravity on cell function, gene expression, and phenotype in Saccharomyces cerevisiae." Appl Environ Microbiol. 2006 Jul;72(7):4569-75. PMID: 16820445 , Jul-2006